SignalAlign

USER MANUAL

Robust delay, coherence and phase analysis

SignalAlign 0.2.28 | English

SignalAlign measures the time difference between related audio signals and reports a robust delay value in samples and milliseconds. It does not move audio by itself; the displayed correction is applied in an external delay, DSP or routing stage.

MEASUREMENT PHILOSOPHY
When the signal is too tonal, periodic or otherwise ambiguous, SignalAlign is designed to withhold an exact delay rather than display a precise-looking but unreliable value.

Developed by	K2consulting
Variants	SignalAlign: VST3 + Standalone | SignalAlign 16: VST3
Signal mapping	REFERENCE = Input 1 | COMPARE = Input 2 or selected Input 2-16

1. Quick Start & Contents

Quick start

1. Route the fixed REFERENCE signal to Input 1.
2. Route the signal to be compared to Input 2. In SignalAlign 16, select one COMPARE input from Inputs 2-16.
3. Leave FREEZE off. SignalAlign measures continuously; there is no AUTO FIND switch in the current version.
4. Start with the factory defaults. For normal operation, SETTINGS exposes FFT Size, Search, Smooth, Low/High and Freeze.
5. Observe the quality panel. A valid delay is released only when the currently applicable reliability checks are satisfied.
6. Read the delay in samples and milliseconds. Apply the indicated correction in an external delay, routing or DSP stage.
7. Measure again after correction. A result near 0 ms confirms alignment.

CURRENT FACTORY DEFAULTS

FFT 32768 | Search +/-500 ms | Smooth Off | Low 50 Hz | High 15000 Hz | Overlap 75% | Averaging Medium | Min Confidence 10% | Min Coherence 15% | Min Peak Ratio 1.00 | Max Phase Diff 0.0 ms | Max Stability 0.00 ms.

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2. Purpose, Signal Flow & Delay Sign

SignalAlign is a measurement tool for two time-related audio signals. Typical applications include microphone alignment, path comparison, broadcast and production signal chains, DSP latency checks and transport-delay verification.

REFERENCE	SignalAlign	COMPARE
Input 1	Analysis	Input 2 / selected 2-16

Delay sign

Delay > 0	Delay < 0
COMPARE arrives later than REFERENCE. To align them by delaying the earlier path, increase delay on REFERENCE by the measured amount.	COMPARE arrives earlier than REFERENCE. Increase delay on COMPARE by the absolute value of the measured amount.

CORRECTION SYMMETRY
REFERENCE correction = measured delay. COMPARE correction = - measured delay. SignalAlign shows both equivalent correction directions above the alignment bar.

Samples and milliseconds

The same delay is displayed in samples and milliseconds. The sample-to-time conversion depends on sample rate.

Sample rate	1 sample	Example: 73 samples
48 kHz	0.020833 ms	1.5208 ms
96 kHz	0.010417 ms	0.7604 ms

Audio pass-through

SignalAlign is an analyser. It passes audio through unchanged. The required timing correction must be applied in an external audio delay, routing stage or DSP.

Variants

Variant	Inputs	REFERENCE	COMPARE	Format
SignalAlign	2	Input 1 / Left	Input 2 / Right	VST3 + Standalone
SignalAlign 16	16	Input 1	Selectable Input 2-16	VST3

3. Main Display & Quality Indicators

The main display combines the accepted delay with signal levels, robustness indicators, correction guidance and frequency-domain diagnostics.

Indicator	Meaning
Delay	Final accepted delay in samples and milliseconds. Positive values mean COMPARE is later.
Confidence	Internal combined reliability score. It is not a standardized measurement quantity.
Coherence	Frequency-domain measure of the linear relationship between REFERENCE and COMPARE.
Peak Ratio	Ratio between the strongest GCC peak and the next independent competing peak.
Stability	Robust variation of raw delay candidates over about one second, before display smoothing.
Phase Fit	Independent delay estimate from the slope of cross-phase across adjacent FFT bins.
Phase Check	Compares broadband GCC delay with Phase Fit when the phase estimate is itself reliable.
Band Check	Soft plausibility check across several frequency regions. It never selects the final delay.
Polarity	Indicates normal or inverted polarity relationship.

Why can the delay disappear?

A current delay is shown only when the applicable quality and ambiguity checks accept the frame. Rejected frames do not enter delay smoothing.

- Confidence is below Min Confidence.
- Coherence is below Min Coherence.
- Peak Ratio is below Min Peak Ratio.
- Phase difference exceeds Max Phase Diff when that gate is enabled and the Phase Check is reliable.
- Stability exceeds Max Stability when that gate is enabled.
- Continuity Tracking has not yet confirmed a remote new peak.
- The material is too tonal, periodic or spectrally sparse for an unambiguous absolute delay.

Last Valid hold

If a trustworthy delay has already been established and the current frame becomes invalid, SignalAlign retains the previous accepted Samples/ms value in grey and labels it LAST VALID | NO CURRENT RESULT. This is display-only: stale values do not drive smoothing or the live alignment bar. Before the first accepted result, the display reads WAITING FOR FIRST VALID RESULT.

4. SETTINGS and ADVANCED

Version 0.2.28 separates everyday controls from engineering and robustness controls. SETTINGS is intended for routine operation; ADVANCED is hidden by default and exposes parameters that are rarely changed.

SETTINGS - day-to-day controls

Parameter	Range / choices	Default	Purpose
FFT SIZE	2048 / 4096 / 8192 / 16384 / 32768 / 65536 / 131072	32768	Analysis window and maximum unambiguous GCC lag.
SEARCH	+/-5 / 10 / 25 / 50 / 100 / 250 / 500 / 750 / 1000 ms	+/-500 ms	Limits the lag region searched for the GCC peak.
SMOOTH	Off / 0.25 / 0.5 / 1 / 2 / 5 s	Off	True output/display time constant after a delay has been accepted.
LOW	20 Hz - 5 kHz	50 Hz	Lower edge of the analysis band.
HIGH	1 kHz - 22 kHz	15000 Hz	Upper edge of the analysis band.
FREEZE	Off / On	Off	Holds the current analysis display. With Freeze off, measurement is continuous.

ADVANCED - robustness controls

Parameter	Range / choices	Default	Purpose
OVERLAP	0 / 50 / 75 / 87.5%	75%	Overlap between consecutive FFT windows.
AVERAGING	Fast / Medium / Slow / Very Slow	Medium	Averages coherence and phase spectra; independent from delay smoothing.
MIN CONFIDENCE	0 - 100%	10%	Minimum internal reliability score.
MIN COHERENCE	0 - 100%	15%	Minimum accepted signal relationship.
MIN PEAK RATIO	1.00 - 3.00	1.00	Minimum dominance of the best GCC peak over the next independent peak.
MAX PHASE DIFF	0.0 - 20.0 ms	0.0 ms	Maximum GCC vs Phase Fit difference. 0.0 disables the hard gate.
MAX STABILITY	0.00 - 2.00 ms	0.00 ms	Maximum raw-delay variation. 0.00 disables the hard gate.

PRACTICAL RECOMMENDATION

Leave ADVANCED at its defaults unless a specific measurement problem requires stricter gating. For most work, Search, FFT Size and the analysis band have the largest practical impact.

FFT size and usable delay range

For a centered N-point GCC, the maximum unambiguous lag is approximately (N/2 - 2) / sample rate. The actually usable range is the smaller of this FFT limit and the selected SEARCH range.

FFT size	Approx. max lag @ 48 kHz	Approx. max lag @ 96 kHz
2048	+/-21.3 ms	+/-10.6 ms
4096	+/-42.6 ms	+/-21.3 ms
8192	+/-85.3 ms	+/-42.6 ms
16384	+/-170.6 ms	+/-85.3 ms
32768	+/-341.3 ms	+/-170.6 ms
65536	+/-682.6 ms	+/-341.3 ms
131072	+/-1365.3 ms	+/-682.6 ms

Practical implications

- With FFT 32768 at 48 kHz, the FFT itself limits the usable lag to about +/-341 ms even if SEARCH is set to +/-500 ms.
- FFT 65536 is sufficient for a full +/-500 ms search at 48 kHz.
- FFT 131072 reaches about +/-1365 ms at 48 kHz, but only about +/-683 ms at 96 kHz; the selected SEARCH range can never exceed the FFT lag limit.

The FFT SIZE tooltip in SignalAlign reports the maximum unambiguous delay for the current sample rate.

Parameter interactions at a glance

Several controls are intentionally coupled. The points below describe the effective behaviour of the current analyser and help avoid changing one parameter while expecting another part of the measurement to remain unchanged.

FFT / SEARCH / sample rate - FFT Size and sample rate set the hard maximum lag; SEARCH can only narrow it. Wider SEARCH does not wait for more audio, adds only a small peak-scan cost, but exposes more competing peaks.

FFT / OVERLAP - Together they set hop size and update rate. Larger FFT updates more slowly; more Overlap updates more often and costs more CPU.

AVERAGING - Averaging is per analysis frame, not a fixed number of seconds. Its real-time settling therefore changes with FFT Size and Overlap.

LOW / HIGH - Bandwidth affects GCC, coherence, phase and ambiguity checks. Narrow or tonal bands carry less unique timing information; HIGH is effectively limited by Nyquist.

CONFIDENCE / gates - Confidence already contains Coherence, Peak Ratio, spectral diversity and supporting checks. Min Coherence, Min Peak Ratio, Max Phase Diff and Max Stability are additional hard gates.

PEAK / PHASE / STABILITY - Min Peak Ratio also influences ambiguity/continuity logic. Max Phase Diff = 0 and Max Stability = 0 disable their hard gates; Phase Diff is only applied when Phase Fit is reliable.

SMOOTH / resets - SMOOTH is output-only and lower Confidence slows convergence somewhat. Changing FFT Size clears averaging/smoothing/continuity/stability history; changing COMPARE in SignalAlign 16 also reacquires from scratch. Stability targets about one second but is frame-count limited at extreme update rates.

For routine work, keep ADVANCED at its defaults and adjust FFT Size, SEARCH and the analysis band first. Use the ADVANCED gates mainly when a specific ambiguity or reliability problem has been identified.

5. Measurement Method & Robust Logic

GCC - Generalized Cross-Correlation

GCC searches for the time shift at which REFERENCE and COMPARE match best. The primary broadband correlation peak supplies the main delay candidate.

PHAT - Phase Transform

With GCC-PHAT, the magnitude of the cross-spectrum is largely normalized so that phase relationships influence the delay search more strongly than simple level differences.

> **SIMPLIFIED PHAT WEIGHTING**
> crossSpectrum / |crossSpectrum|

Coherence- and energy-aware broadband GCC

SignalAlign keeps a single broadband GCC-PHAT estimate as the master. Frequency bins with strong coherence contribute more reliably, very weak regions are reduced, and loud isolated tones are prevented from dominating the estimate.

Peak Ratio and ambiguity detection

Periodic or complex material can produce several plausible GCC peaks. Peak Ratio compares the strongest peak with the next independent competitor. Values near 1.0 indicate ambiguity.

Peak Ratio	Interpretation
1.80 : 1	Very clear
1.35 : 1	Good
1.15 : 1	Borderline
1.03 : 1	Nearly ambiguous

Phase Fit and Phase Check

A pure delay produces a frequency-dependent phase slope. SignalAlign derives an independent second delay estimate from adjacent-bin cross-phase and can compare it with GCC. The hard phase-difference gate is only relevant when enabled and when the phase estimate itself is reliable.

Continuity Tracking

Once a stable delay is established, a distant new peak cannot replace it because of a single frame. A true new delay must persist for several consecutive frames. Switching the COMPARE source in SignalAlign 16 resets analysis history so the new source is reacquired cleanly.

Tonality / sparse-spectrum rejection

Strongly tonal or periodic material may have many mathematically equivalent delays. A 1 kHz sine repeats every 1 ms, for example. High coherence alone does not make the absolute delay unique. In such cases SignalAlign can deliberately withhold a delay.

Band Check

Several frequency regions test whether their phase progression is compatible with the broadband GCC delay. A display such as BAND CHECK 5/6 means five of six usable regions agree. Band Check is a soft plausibility factor only.

6. Alignment Bar, Corrections & Display Behaviour

Alignment bar

The horizontal alignment bar is centered at 0 ms. Left means COMPARE is earlier; right means COMPARE is later. Colour evaluates the absolute delay symmetrically.

Region	Factory threshold	Meaning
Green	<= 1.9 ms	Within the green alignment region.
Orange	> 1.9 ms to 7.9 ms	Intermediate deviation.
Red	> 7.9 ms	Large deviation.

Right-click the alignment bar to change the green and red limits. The thresholds are stored with the plug-in state.

Correction arrows

Right arrow - green
Increase delay on the indicated path by the displayed amount.

Left arrow - red
Reduce delay on the indicated path by the displayed amount.

REFERENCE and COMPARE always show opposite correction values. Their amounts follow the smoothed display delay. Last Valid corrections remain visible but are dimmed.

Display precision

Right-click the main Samples or ms readout to select 0-4 decimal places independently. Factory defaults are 0 decimals for Samples and 2 decimals for milliseconds. The millisecond correction readouts follow the selected ms precision.

Smoothing

SMOOTH is output-only. Quality decisions and Stability use raw delay candidates before output smoothing. Smooth Off displays a newly accepted delay immediately; the time-constant options make the displayed result approach a new accepted delay gradually.

Freeze

FREEZE is the deliberate analysis hold. With FREEZE off, SignalAlign measures continuously. Freeze is useful when a transient good measurement needs to be inspected or documented.

7. SignalAlign 16 & REAPER Workflow

SignalAlign 16

SignalAlign 16 is a separate 16-input VST3. Input 1 is always REFERENCE. Inputs 2-16 are selectable COMPARE sources, with exactly one active at a time.

- The 15 COMPARE selectors are exclusive round buttons labelled 2 through 16.
- The active input is highlighted in blue.
- Changing COMPARE resets analysis history and continuity tracking before reacquisition.
- The selected COMPARE input is stored as an automatable plug-in parameter.
- All channels pass through unchanged.

ROUTING NOTE
A host track containing SignalAlign 16 should provide 16 channels if all compare inputs are to be available. Input 1 remains the fixed REFERENCE path.

Optional REAPER Auto Compare Router

An optional REAPER Lua script can follow the selected track and route it as COMPARE while keeping the REFERENCE path fixed. This is useful when many tracks need to be checked against one reference.

1. Load SignalAlign on a dedicated measurement track.
2. Route REFERENCE permanently to the reference input.
3. Start the Auto Compare Router ReaScript.
4. Select a track to be measured.
5. The script routes that track as COMPARE and removes/replaces the previous temporary compare routing.
6. When stopped, the helper routing is removed and temporary track colouring is restored.

IMPORTANT
The compare routing must not accidentally mix with an existing right-channel signal on the measurement track. Use the router/injector arrangement intended for SignalAlign so only the selected compare path reaches the analysis input.

Help, About and manual

Clicking the K2 / SignalAlign brand area opens the integrated help menu in the same dark COMISRatio-style colour scheme as the plug-in. The menu includes a compact Parameter Interactions section, About SignalAlign, the online manual and Third-Party Licenses. The About window uses K2consulting branding, shows the plug-in and JUCE version, and links to the current manual. The permanent manual location is: https://nc-5789769951021120545.nextcloud-ionos.com/index.php/s/WHckLHmraP5fd7b?dir=/&editing=false&openfile=true

Licensing and third-party components

SignalAlign's Third-Party Licenses window lists the framework and SDK notices relevant to the current release build. JUCE is used under the JUCE licence applicable to K2consulting's build and distribution. The current GitHub release build uses JUCE 8.0.15.

JUCE 8.0.15 vendors Steinberg VST 3 SDK 3.8.0. VST 3 SDK 3.8.0 is MIT licensed; the About/Third-Party window reproduces the Steinberg copyright and MIT permission notice. The VST name is a Steinberg trademark and any optional use of VST branding must follow Steinberg's usage guidelines.

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8. Troubleshooting & Interpretation

Observation	Likely cause	Recommendation
Low Coherence	Signals are not sufficiently linearly related; different sources, room contribution or different processing.	Check routing and analysis band; wait for more suitable material.
Peak Ratio near 1.0	Several GCC peaks are almost equally plausible; common with periodic or tonal material.	Use transient/broadband material; narrow SEARCH; do not force a low threshold if reliability matters.
High Stability value	Raw delay candidates are moving from frame to frame.	Narrow SEARCH; review analysis band; avoid periodic material; enable a Stability gate if needed.
Phase Fit far from GCC	Phase slope and correlation peak do not describe the same simple delay.	Check reflections, filters, multipath and routing before relaxing the phase gate.
Weak Band Check	Only a few frequency regions support the broadband delay.	Treat it as a warning; Band Check alone does not select or reject the delay.
TONAL / AMBIGUOUS	Spectrum is too sparse or periodic for unique absolute timing.	Do not force it. Use broader-band or more transient material.
Correct delay but slow display	Smooth, Averaging or FFT window is too slow for the desired response.	Reduce Smooth, use faster Averaging or reduce FFT size.

Why a 1 kHz test tone does not give a unique delay

A 1 kHz sine repeats every 1 ms. Delays separated by integer multiples of 1 ms can therefore produce the same phase relationship. Coherence may be high because the two signals are strongly related, yet the absolute time offset remains ambiguous. Pink noise, transients or normal programme material provide far more unique timing information.

KEY DISTINCTION
Coherence answers whether the signals are related. It does not guarantee that the absolute delay is uniquely identifiable.

Old VST3 copies in REAPER

REAPER scans every file that still ends in .vst3 inside its plug-in paths. A renamed backup such as SignalAlign_old.vst3 can still be detected as a plug-in. Keep backups outside scanned folders or change the suffix, for example to .vst3.backup.

9. Glossary & Reference

Term	Definition
REFERENCE	Fixed reference signal on Input 1.
COMPARE	Signal being compared with REFERENCE; Input 2 in the standard version, selectable Input 2-16 in SignalAlign 16.
Delay	Time difference between REFERENCE and COMPARE.
GCC	Generalized Cross-Correlation; method used to estimate time offset.
PHAT	Phase Transform; normalizes cross-spectrum magnitude so phase has stronger influence on the delay search.
GCC-PHAT	Broadband GCC with PHAT weighting; the primary delay estimator.
Coherence	Measure of the linear relationship between two signals across frequency.
Confidence	SignalAlign's internal combined reliability score.
Peak Ratio	Strongest independent GCC peak divided by the next strongest independent peak.
Phase Fit	Independent delay estimate derived from cross-phase slope.
Phase Check	Plausibility comparison between GCC delay and Phase Fit.
Stability	Robust spread of raw delay candidates in milliseconds, before smoothing.
Continuity Tracking	Protection against isolated jumps to a distant competing peak.
Band Check	Soft multi-band plausibility check.
Tonality Rejection	Detection of strongly periodic or spectrally sparse signals with ambiguous timing.

Reference defaults - SignalAlign 0.2.28

Parameter	Factory default
FFT Size	32768
Search	+/-500 ms
Smooth	Off
Low / High	50 Hz / 15000 Hz
Overlap	75%
Averaging	Medium
Min Confidence	10%
Min Coherence	15%
Min Peak Ratio	1.00
Max Phase Diff	0.0 ms (gate disabled)
Max Stability	0.00 ms (gate disabled)
Alignment thresholds	Green <= 1.9 ms; Red > 7.9 ms
Display precision	Samples 0 decimals; ms 2 decimals